(2) SHELL STRUCTURE OF NUCLEII. MAGIC NUMBERS: Central potential & Spin orbit Coupling.

Though Liquid drop model successfully explain the mass of nucleus, binding energy, fission etc but there are other properties which can not be explained by Liquid drop model.

In order ~~to~~ explain few properties of Nucleus ~~for which~~ we have to imagine that there exist Shells in the nucleus where proton and neutron can be accomodated similar to the electron Shells outside nucleus. When a shell is completed by electron the last electron possesses maximum binding energy (Ionization energy).

(b). The elements having proton number equal to the magic number have greater number of Stable Isotopes than the neighbouring atom.
e.g: For tin ($Z = 50$) the Stable Isotopes are 10. where as for Cadmium ($Z = 48$) or Telurium ($Z = 52$) have 8 Stable Isotopes.

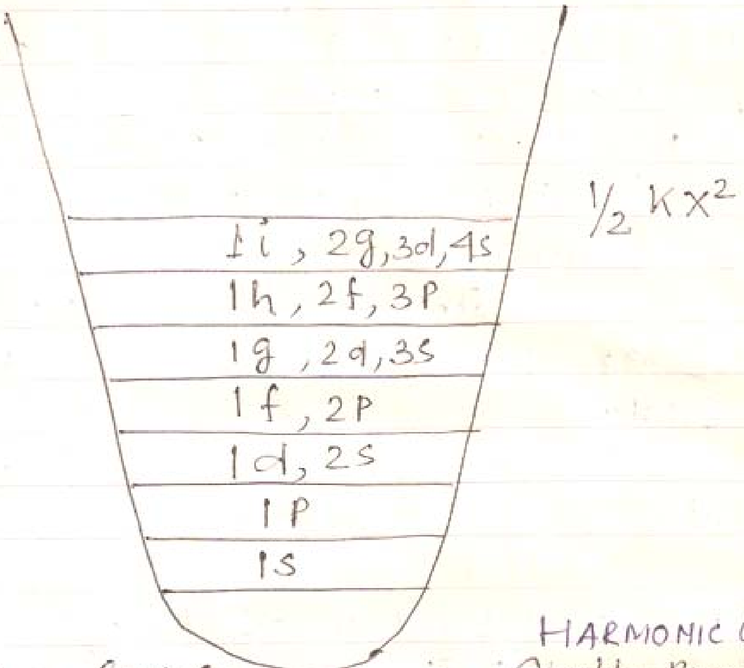
Similarly elements having neutron number equal to magic no. have ~~greater~~ large no. of Isotones.
e.g for $N = 82$ the no. of Iso Stable Isotone $= 7$. but for $N = 80$ or $N = 84$ no. of Stable Isotones 3 & 2 respectively.

(c). The end product of three Natural radioactive series containing magic number of protons $Z = 82$ is the Isotope of Lead.

(d) The nucleus containing magic number of Proton or Neutron have much higher value for its 1st excited energy state in comparision to the neighbouring Nucleus.

(e) The nucleus containing magic number of proton or Neutron has smaller Neutron Capture cross-section. This is because that since the nucleus shell is completed therefore probability of proton or Neutron Capture is much less.

According to Quantum mechanics when a physical System such as atomic electron system or Nucleus system is contained in a potential well then that physical system can exist in definite energy state.



Nucleus energy states in ~~Simple Resonance~~ HARMONIC OSCILLATOR Potential

FIG : 1

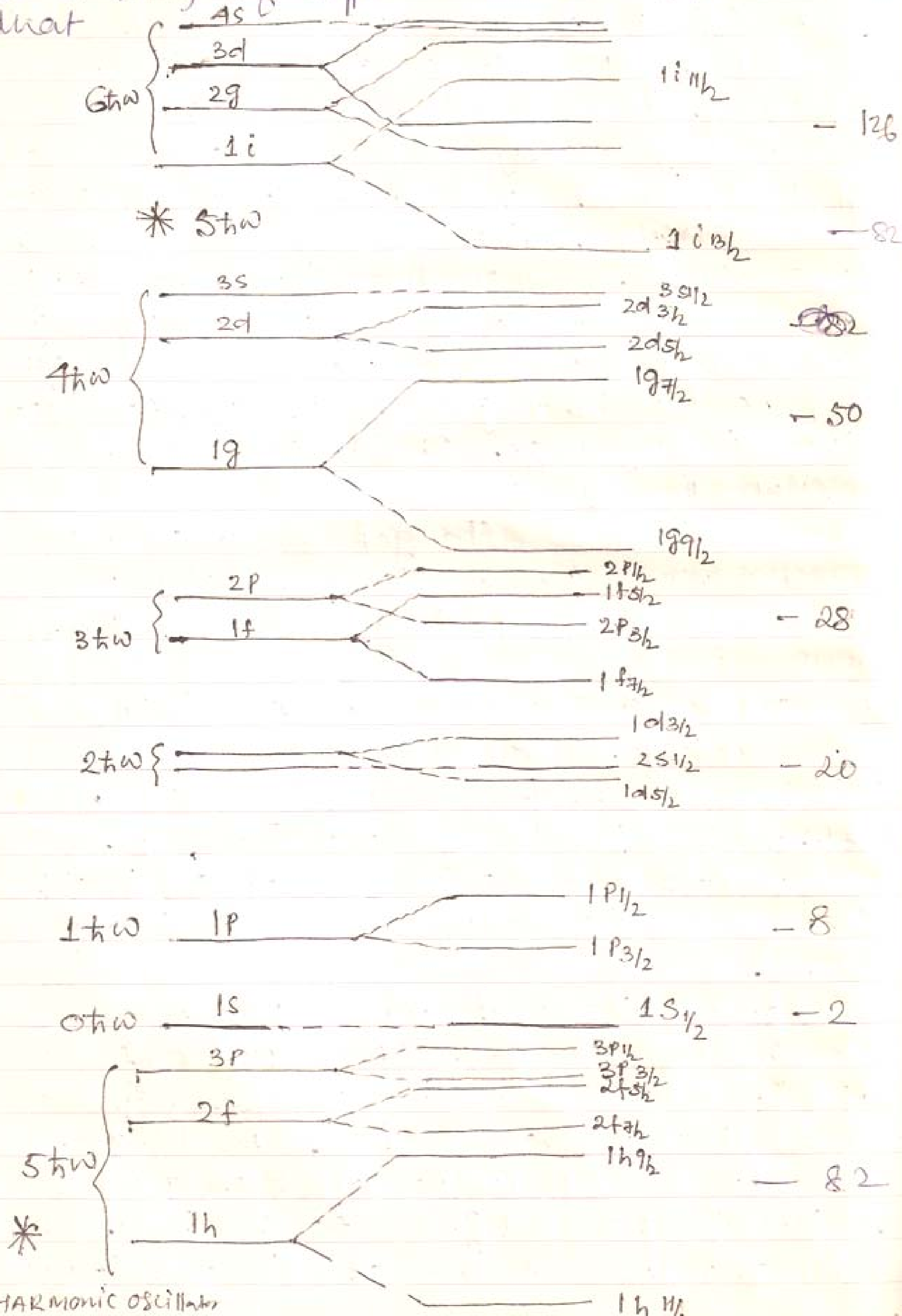
Mayer and Haxel assumed that Protons and Neutrons exists in a Central potential well. But this is not Static electric potential field. Protons and Neutrons exist in a average potential well by the mutual attraction of particles. In the above average potential well every nucleon moves independently. By putting the mathematical expression in Schrödinger wave equation the probable energy states of Nucleons in nucleus can be Obtained. Pauli's exclusion principle determines the existence of Nucleons in different energy states. Similar to the atomic System in this Case also Nucleons are filled up begining from the lower energy States and then to the respective higher energy states. Generally by assuming infinite potential well or Harmonic Oscillator potential well the 3-dimensional Schrödinger wave eqn is solved. Using this theory the existence of Nuclear energy shells can be explained. It is found few energy states exist very closely and then after a wide gap again energy states exists closely FIG: 2. The closely situated energy states belongs to a Shell. whenever a Shell is completed with even number of nucleon then the energy needed to remove a nucleon is much higher. Thus a the binding energy of completely filled Shell is very large. The nucleon no. of such well. Shell is a magic number.

To get the above magic numbers we have to assume that the Spin-orbit Coupling force, is much higher.

Due to L-S Coupling the orbital angular moment lth Splits into two Sublevels. The total angular momentum Quantum number for the two Sublevels is respectively $J = l + \frac{1}{2}$ and $J = l - 1/2$. Only when $l = 0$ then one value of $J = 1/2$ is possible. Since l is an Integer hence J is a fractional number.

Theoritically $J = l + 1/2$ state posses lower energy than $J = l - 1/2$ State. The difference between the energy states increases with the increase in the value of l. For $l > 3$ the difference is so large that these two Sublevels exists in two different energy Shells. The total angular momentum of the nucleus (or Nuclear Spin)

Quantum number I is determined by the J-vector coupling of different Nucleons. It is evident that



(HARMONIC oscillator level)

(splitting due to Spin orbit force)

FIG: 2 Effect of spin orbit coupling on the level system of a well of a shape intermediate between the square and oscillator wells. The magic numbers are given on far right.

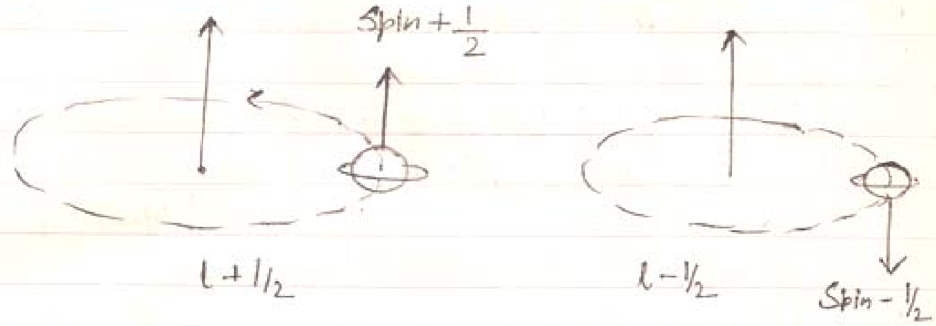


FIG: 3

Coupling of orbital and spin angular momenta of a nucleon.

According to the Quantum no. n, l, J different nuclear energy states are represented.

$1S_{1/2}$, $1P_{3/2}$, $1P_{1/2}$, $1d_{5/2}$, $1d_{3/2}$, $2S_{1/2}$ ---- $2f_{7/2}$ etc

According to the Pauli's exclusion principle for energy sublevel J there can be $(2J+1)$ number of Nucleons (Proton or Neutron). When the sublevel is filled by $(2J+1)$ number of like nucleon then the excess like nucleons shifts to the higher energy state.

In the figure according to the Nuclear shell model ~~the~~ different probable Nuclear energy states have been ~~are~~ shown. The number of like nucleons that are fills the shell is also shown.

From the fig: 2 the importance of L-S coupling for the nuclear energy state splitting can be understood.

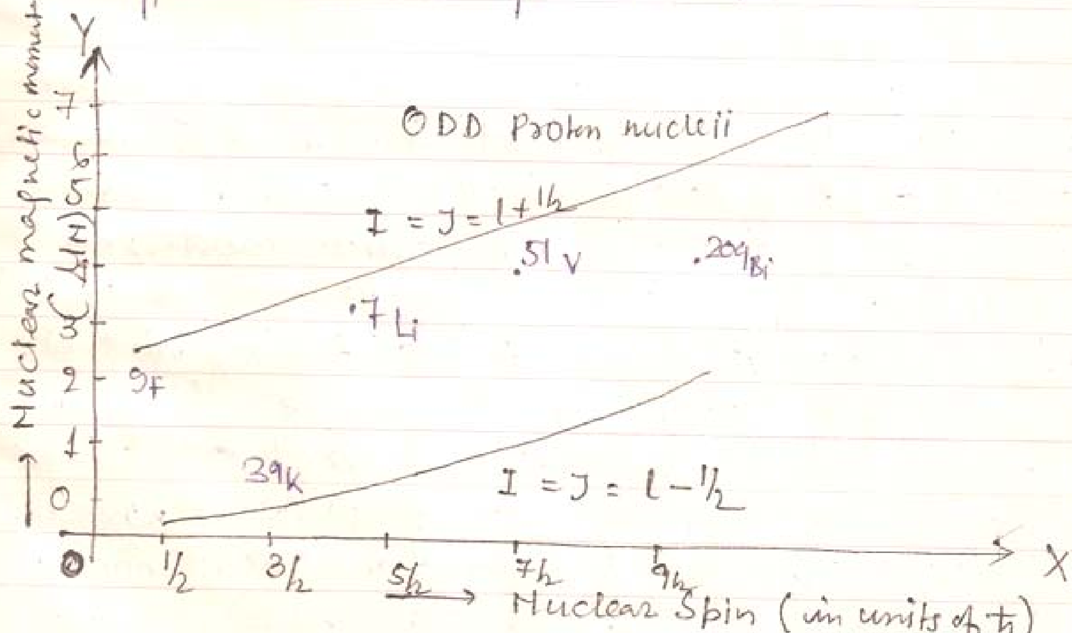
e.g. the energy of the sublevel $1f_{5/2}$ is ~~so~~ ~~high~~ larger than $1f_{7/2}$ $(l=3)$ ~~that~~ Hence as soon as the last sublevel is filled up the whole shell got completely filled

Consequently magic number 28 arises. Similarly the energy difference of $1g_{9/2}$ and $1g_{7/2}$ $(l=4)$ is so high that they are situated in different shells for which magic number 50 arises.
Similarly magic number 82 and 126 arises due to L-S spliting of the sublevels $1h_{11/2}$ & $1h_{9/2}$ $(l=5)$ and $1i_{13/2}$, $1i_{11/2}$ $(l=6)$ respectively.
using ~~Nuclear~~ Shell model of Nucleus not only magic number but also other properties such as Nuclei Spin, magnetic moment etc can be explained. To explain we assume:
For a definite J the like even nucleons (Proton or Neutron) are arranged in such a way that the total angular momentum or spin becomes zero. Hence for even numbered nucleons angular momentum is zero. On the other hand for the odd numbered Nucleon shell the angular momentum is equal to the angular momentum of the last unpaired electron. Thus we conclude

When $Z =$ Even, if $N =$ even then Nuclear Spin $I = 0$
when $Z =$ even ; $N =$ Odd then I is determined by the J of last odd numbered Neutron.
when $Z =$ odd, $N =$ even, then I is determined by the J of last odd numbered Proton

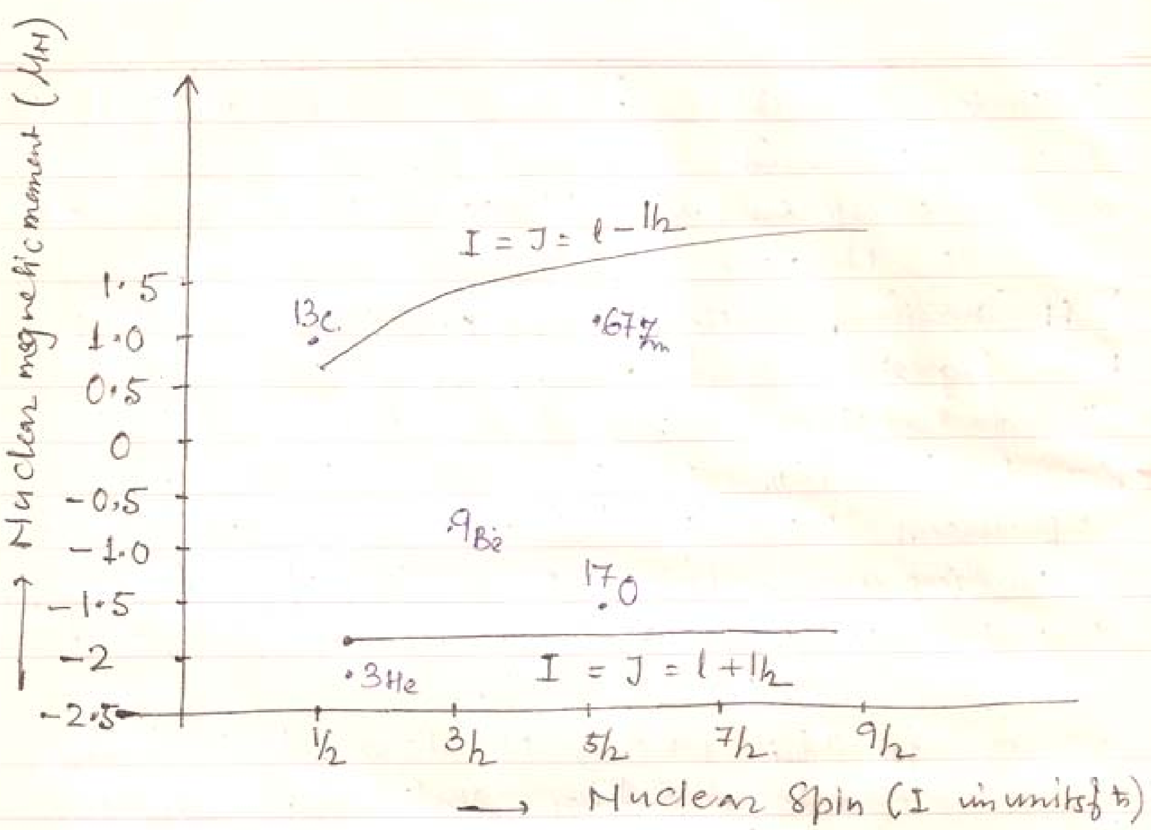
Based on the above assumptions the nuclear shell model is termed as single particle model. using this model the Nucleus possessing Even-Even & odd A ; the Nuclear Spin can be determined.

We know that for Even-Even & odd A the Nuclear Spin is zero. ($I=0$).
let us Consider the Nuclii having odd A.
For the Nucleus
^3H ($Z=1$) & ^3He ($N=1$) we get $I = \frac{1}{2}$
from fig ② we find for both Cases the Odd nucleon (respectively 1 proton & 1 Neutron) exist in $1S_{1/2}$ energy state. Since for each of them having like even nucleons (respectively 2 Neutrons and 2 proton) $I=0$ Therefore for both Nucleus the Nuclear Spin is equal to the last odd Nucleon $J = \frac{1}{2}$ i.e $I = \frac{1}{2}$.
Again for ^7Li ($Z=3$) Nucleus the 3rd proton exist in $1P_{3/2}$ state because the lowest state $1S_{1/2}$ is filled up by $2J+1=2$ protons. Hence in this case $I = \frac{3}{2}$ which agrees with experimental result. Actually this model successfully explain the I for different Nucleus for 3 odd A Nucleus.



Schmidt lines (for odd Proton nucleus).

FIG: 4.



Schmidt lines for Nucleus having odd Neu

using Shell model — the value of magnetic moment for different nucleus having odd A Can be found.

The observed nuclear magnetic moments can also be understood in terms of J value of the odd nucleon in the nucleus. On the basis of the independent particle model, we expect the spins and magnetic moments of Isotopic nucleii of odd A, differing in their A values by two, not to change, since the two added Neutrons are paired off and do not affect the single odd Nucleon which is responsible for the spin and magnetic moment of the Isotopic nuclei. This is found to be the case with odd A Isotopes.

Magnetic moment of Single odd Nucleon $= (g_l l^* + g_s s^*)\mu_N$ —(1)

where g_l and g_s are respectively are orbital & Spin 'g' factor of the single odd nucleon &

$$\vec{j^*} = \vec{l^*} + \vec{s^*}, \quad \vec{I} = j$$

The value μ of the nuclear magnetic moment experimentally obtained in the case where $I = j$ is

$$\mu = (\text{the nucleon magnetic moment along } \vec{j^*}) \frac{j}{j^*}$$

$$= [g_l l^* \cos(\vec{l^*} \vec{j^*}) + g_s s^* \cos(s^* j^*)] \frac{j}{j^*}\mu_N \quad \text{—(2)}$$

The Cosines are evaluated from:

$$\left.\begin{aligned} s^{*2} &= j^{*2} + l^{*2} - 2 j^* l^* \cos(\vec{l^*} \vec{j^*}) \\ \& \quad l^{*2} &= j^{*2} + s^{*2} - 2 j^* s^* \cos(\vec{s^*} \vec{j^*}) \end{aligned}\right\}$$

$\therefore$ from (2):-

$$\mu = \left(g_l \frac{j^{*2} + l^{*2} - s^{*2}}{2j^*} + g_s \frac{j^{*2} + s^{*2} - l^{*2}}{2j^*}\right)\frac{j}{j^*}\mu_N$$

taking $S = 1/2$ for a Single Nucleon we get

$$\left.\begin{aligned} &\text{For } j = l + 1/2 \;;\; l = j - 1/2 \;,\; \mu = [(j - 1/2)g_l + \tfrac{1}{2}g_s]\mu_N \\ &\text{For } j = l - 1/2 \;;\; l = j + 1/2 \;;\; \mu = \frac{j}{j+1}[(j + 3/2)g_l - \tfrac{1}{2}g_s]\mu_N \end{aligned}\right\} \text{—(3)}$$

The μ versus j curves gives Schmidt lines. It is found that (μ, j) points do not fall on the Schmidt lines rather confined to the region between the Schmidt lines.

The observed anomalous magnetic moments of nucleons may be attributed to their basic Structure of a nucleonic core Surrounded by the presence of the other nucleons in a nucleus, the magnetic moment of a nucleon in a nucleus may not be same as that of the free nucleon. This may probably explain the failure of the shell model to predict quantitatively the magnetic.